Understanding Systems and Their Boundaries

- To model a system effectively, it's important to grasp what a system is and where its limits lie.
- A system is basically a group of connected parts working together for a common goal.
- For example, think of a car factory where machines, parts, and workers all work together to make cars.
- Systems are influenced by things outside of them, which we call the ‘system environment’.
- These external factors can affect how the system works. When we model systems, we have to decide what's part of the system and what's part of the environment. This depends on what we're trying to figure out.
- For instance, in a factory, things like order arrivals might be considered outside the factory's control. But if supply and demand affect how much the factory produces, that relationship becomes part of the system.
- Similarly, in a bank, there might be rules about how much interest can be offered. When we study one bank, these rules are just part of the environment. But if we're looking at how banking laws affect the whole industry, those rules become part of the system we're studying.
- So, understanding systems and their boundaries helps us figure out which parts and interactions to include in our models and which ones to consider external factors.

Components of a System

1. Entity: An entity is something of interest within the system. For example, in a bank system, customers would be entities.
2. Attribute: An attribute is a characteristic or property of an entity. For instance, the balance in a customer's checking account is an attribute.
3. Activity: An activity represents a specific period of time within the system. For example, making deposits at a bank would be considered an activity.

- When studying a system, the collection of entities involved might differ depending on the focus of the study.
- For example, if we're looking at how many tellers a bank needs, the system might include regular tellers and customers waiting in line. But if we're also considering special tellers for specific transactions, we'd need to expand the system definition.
- The state of a system refers to the variables needed to describe it at any given time, based on the study's objectives.
- For instance, in our bank study, variables like the number of busy tellers, customers waiting, and the next customer's arrival time are important state variables.
- An event is an instantaneous happening that could change the system's state. For example, a customer arriving at the bank is an event that affects the system's state.
- Endogenous activities and events occur within the system itself, while exogenous ones happen in the environment but impact the system. For instance, a customer arriving (exogenous) and a customer completing a transaction (endogenous) in the bank study.
- The table below describes a couple of systems by its components. Note that the list of components is not exhaustive:

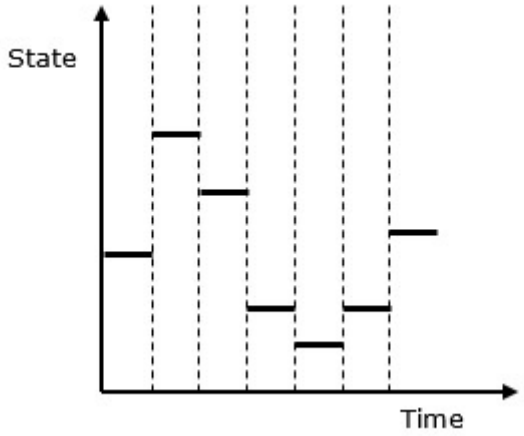
System	Entities	Attributes	Activities	Events	State
Banking	Customers	Checking account balance	Making deposits	Customer arrival, departure	Number of busy tellers, number of customers waiting
Communication	Messages	length	Transmitting	Arrival of message, sending of message	Number of messages in que,
Inventory	Warehouse	Capacity	Product delivery,	Demand	Inventory level; backlogged

			withdrawing		demands

Discrete and Continuous Systems

- Systems can be classified as either discrete or continuous. While real-world systems may exhibit aspects of both types, they tend to lean towards one predominant type.

1. Discrete Systems: In a discrete system, the state variables change only at specific points in time. For example, consider a bank where the number of customers changes only when someone arrives or completes a transaction. This discrete change can be visualised as a step function, where the state variable jumps from one value to another at discrete time intervals.



2. Continuous Systems: In contrast, a continuous system experiences continuous changes in its state variables over time. An example is the head of water behind a dam. Here, the water level fluctuates continuously due to factors such as rainfall, water draw for flood control and electricity generation, and evaporation. This continuous change is represented by a smooth, continuous curve, showing how the state variable changes steadily over time.

Modelling a System

- Studying a system often involves understanding how its components interact or predicting its behaviour under different conditions.
- While experimenting directly with the system is sometimes possible, it's not always practical or feasible.
- For instance, testing a new policy in a bank by reducing teller numbers might upset customers. In such cases, studies are conducted using models of the system.
- A model is essentially a representation of a system used for studying it. It focuses on the aspects of the system relevant to the problem at hand.
- While a model simplifies the system, it must still capture enough detail to draw valid conclusions about the real system.
- Different models may be needed for different study purposes.
- Similar to the components of a system, a model includes entities, attributes, and activities.
- However, the model only includes those components essential to the study. This allows researchers to analyse and understand the system's behaviour without unnecessary complexity.

Types of Models

Models can be categorised into various types, each serving different purposes and representing systems in unique ways.

1. Form of Representation:
    - Physical Models: Represent systems through physical replicas or scaled-down versions of objects.
    - Mathematical Models: Use mathematical equations and symbolic notation to represent systems.

2. Nature of Variation:
    - Static Models: Depict systems at specific points in time without considering changes over time.
    - Dynamic Models: Represent systems as they evolve and change over time.

3. Incorporation of Randomness:
    - Deterministic Models: Have fixed inputs leading to predictable outputs without randomness.
    - Stochastic Models: Incorporate random variables as inputs, leading to random outputs.

4. Nature of Time:
    - Discrete Models: Describe systems where state variables change only at discrete points in time.
    - Continuous Models: Represent systems with continuous changes in state variables over time.

- Mixed models combine elements of discrete and continuous models based on system characteristics and study objectives.